# Math 372: Homework 04

*Due Wednesday, Feb 11*

**Problem 1** (Exercise 2.3.34)**.** Computer keyboard failures can be attributed to electrical defects or mechanical defects. A repair facility currently has 25 failed keyboards, 6 of which have electrical defects and 19 of which have mechanical defects.

(a) How many ways are there to randomly select 5 keyboards for a thorough inspection (without regard to order)?

(b) In how many ways can a sample of 5 keyboards be selected so that exactly 2 have an electrical defect?

(c) If a sample of 5 keyboards is randomly selected, what is the probability that at least 4 of them will have a mechanical defect?

**Problem 2** (Exercise 2.3.39)**.** A box in a supply room contains 15 compact fluorescent lightbulbs, of which 5 are rated 13-watt, 6 are rated 18-watt, and 4 are rated 23-watt. Suppose that three of these bulbs are randomly selected.

(a) What is the probability that exactly two of the selected bulbs are rated 23-watt?

(b) What is the probability that all three of the bulbs have the same rating?

(c) What is the probability that one bulb of each type is selected?

(d) If bulbs are selected one by one until a 23-watt bulb is obtained, what is the probability that it is necessary to examine at least 6 bulbs?

**Problem 3** (Exercise 2.4.47)**.** Consider randomly selecting a student at a large university. Let $A$ be the event that the selected student has a Visa card, let $B$ be the event that the student has a MasterCard, and let $C$ be the event that they have an American Express card. Suppose that

$$\mathbb{P}[A] = .6, \quad \mathbb{P}[B] = .4, \quad \mathbb{P}[C] = .2 \tag{1}$$
$$\mathbb{P}[A \cap B] = .3, \quad \mathbb{P}[A \cap C] = .15, \quad \mathbb{P}[B \cap C] = .1, \tag{2}$$
$$\text{and} \quad \mathbb{P}[A \cap B \cap C] = .08. \tag{3}$$

(a) What is the probability that the selected student has at least one of the three types of cards?

(b) What is the probability that the selected student has both a Visa card and a MasterCard, but not an American Express card?

(c) Calculate and interpret $\mathbb{P}[B \mid A]$ and $\mathbb{P}[A \mid B]$.

(d) If we learn that the selected student has an American Express card, what is the probability that she or he also has both a Visa card and a MasterCard?

(e) Given that the selected student has an American Express card, what is the probability that she or he has at leaset one of the other two types of cards?

**Problem 4** (2.4.63)**.** For customers purchasing a refrigerator at a certain appliance store, let $A$ be the event that the refrigerator was manufactured in the U.S., $B$ be the event that the refrigerator had an icemaker, and $C$ be the event that the customer purchased an extended warranty.

Relevant probabilities are

$$\mathbb{P}[A] = .75 \quad \mathbb{P}[B \mid A] = .9 \quad \mathbb{P}[B \mid A^c] = .8$$
$$\mathbb{P}[C \mid A \cap B] = .8 \quad \mathbb{P}[C \mid A \cap B^c] = .6$$
$$\mathbb{P}[C \mid A^c \cap B] = .7 \quad \mathbb{P}[C \mid A^c \cap B^c] = .3.$$

(a) Construct a tree diagram consisting of first-, second-, and third-generation branches, and place an event label and appropriate probability next to each branch.

(b) Compute $\mathbb{P}[A \cap B \cap C]$.

(c) Compute $\mathbb{P}[B \cap C]$.

(d) Compute $\mathbb{P}[C]$.

(e) Compute $\mathbb{P}[A \mid B \cap C]$, the probability of a U.S. purchase given that an icemaker and extended warranty are also purchased.

**Problem 5** (Exercise 102 on p.93)**.** A sample of 2026 students were asked about their eye color, with the following results:

| | **Blue** | **Brown** | **Green** | **Hazel** | **Total** |
|---|---|---|---|---|---|
| **Male** | 370 | 352 | 198 | 187 | 1107 |
| **Female** | 359 | 290 | 110 | 160 | 919 |
| **Total** | 729 | 642 | 308 | 347 | 2026 |

Suppose that one of these 2026 students is randomly selected. Let $F$ denote the event that the selected individual is female, and $A, B, C$ and $D$ represent the events that they have blue, brown, green, and hazel eyes, respectively.

(a) Calculate $\mathbb{P}[F]$ and $\mathbb{P}[C]$

(b) Calculate $\mathbb{P}[F \cap C]$. Are these events independent? Why or why not?

(c) If the selected individual has green eyes, what's the probability that he or she is a female?

(d) If the selected individual is female, what is the probability that she has green eyes?

(e) What is the "conditional distribution" of eye color for females? (i.e., $\mathbb{P}[A|F], \mathbb{P}[B|F], \mathbb{P}[C|F], \mathbb{P}[D|F]$), and what is it for males? Compare the two distributions.

**Problem 6.** Two cards are randomly drawn from a deck of cards. Let

$$\begin{aligned} A &= [\text{at least one ace is drawn}] \\ B &= [\text{both cards are aces}] \\ S &= [\text{the ace of spades is drawn}] \end{aligned}$$

Compute the following conditional probabilities:

(a) $\mathbb{P}[S \mid B]$

(b) $\mathbb{P}[B \mid S]$

(c) $\mathbb{P}[B \mid A]$

(d) Note that the answer to (b) is almost twice as large as the answer to (c). Do you find this result surprising?

**Problem 7.** Recall the following definition:

$$\boxed{\text{Events } A \text{ and } B \text{ are } \textbf{\textit{independent}} \text{ if and only if } \mathbb{P}[A \cap B] = \mathbb{P}[A]\,\mathbb{P}[B]}$$

Assume that $A$ and $B$ are independent. Explain way $A^c$ and $B$ must be independent, and also why $A^c$ and $B^c$ must be independent.

**Problem 8** (Exercise 2.5.71)**.** A space mining company currently has two active projects, one on the moon and one in the Kuiper belt. Let $L$ be the event that the lunar mining project is successful, and let $K$ be the event that the Kuiper belt mining project is successful. Suppose that $L$ and $K$ are independent events with $\mathbb{P}[L] = 0.4$ and $\mathbb{P}[K] = 0.7$.

(a) If the lunar project is not successful, what is the probability that the Kuiper belt project is also not successful? Explain your reasoning.

(b) What is the probability that at least one of the two mining projects will be successful?

(c) Given that at least one of the two projects is successful, what is the probability that only the lunar project is successful?

**Problem 9** (Exercise 2.5.77)**.** Five explosive devices are placed underneath the walls of the enemy stronghold, and are set to detonate simultaneously. Each bomb has a 4% chance of failing to detonate at the specified time. Assuming that the probabilities are independent, calculate $\mathbb{P}[\text{at least one detonation occurs}]$ and $\mathbb{P}[\text{at least one bomb fails to detonate}]$

**Problem 10** (Similar to Example 2.31)**.** An email provider observes that 50% of emails are spam. It intends to filter spam emails before they reach the inbox. The filter's accuracy for detecting spam email is 99% and chances of tagging a non-spam email as spam is 5%.

(a) Draw a tree diagram to illustrate this situation

(b) If a certain email is tagged as spam, find the probability that it is not a spam email. That is, find

$$\mathbb{P}[\text{email is not spam} \mid \text{email is tagged as spam}]$$

**Problem 11.** Consider a universe where Netflix has exactly 6 shows, numbered 1 through 6. Suppose we randomly select a subscriber and ask which shows they watch. For each $i = 1, 2, \ldots, 6$, let $E_i = \{i\}$ be the event that the subscriber watches show $i$. Suppose that

$$P(E_1) = P(E_6) = .10, \quad P(E_2) = P(E_5) = .15, \quad P(E_3) = P(E_4) = .25.$$

Define events $A, B, C$ by

$$A = \{2, 4, 6\}, \quad B = \{1, 2, 3\}, \quad C = \{2, 3, 4, 5\}, \quad D = \{1, 3, 5\}.$$

(a) Are $A$ and $B$ dependent or independent?

(b) Are $A$ and $C$ dependent or independent?

(c) Are $A$ and $D$ dependent?

(d) Can two mutually exclusive events $A$ and $B$ with $P(A) > 0$ and $P(B) > 0$ be independent? Why or why not?

**Problem 12.** The prevalence of bird flu in a wild bird population is 1%. A new diagnostic test is advertised as having a false negative rate of 0.03 and a false positive rate of 0.05. To be precise, this means that

$$\mathbb{P}[\text{test is negative} \mid \text{bird has the flu}] = 0.03$$

$$\mathbb{P}[\text{test is positive} \mid \text{bird doesn't have the flu}] = 0.05$$

(a) What is the probability that a randomly-selected bird tests positive? *(You don't need to simplify your answer.)*

(b) Suppose that a randomly-selected bird tests positive. What is the probability that the bird has the disease? *(You don't need to simplify your answer.)*

**Problem 13** (Rolling two dice)**.** Two 6-sided dice are rolled and added together. Let $X$ be the sum.

(a) What is the probability mass function of $X$? [*Hint:* First determine $p(1), p(2), \ldots$ and so on.]

(b) Determine the cumulative density function of $X$ and graph it.

(c) What is the expected value of $X$?

**Problem 14** (Rolling a d6 'with advantage')**.** In the game *Dungeons & Dragons*, there is a dice-rolling mechanic called "rolling with advantage" in which the player gets to roll a dice twice and keep the largest value. This bonus is typically regarded as a good thing, as higher rolls tend to be better in the game. In this problem, we'll investigate how significant this bonus is. Two 6-sided dice are rolled. Let $M$ be the maximum of the two rolls.

(a) What is the probability mass function of $M$? [*Hint:* First determine $p(1), p(2), \ldots$ and so on.]

(b) Determine the cumulative density function of $M$ and graph it.

(c) What is the expected value of $M$?

(d) Suppose you are given the choice between two magical weapons: a magical sword that deals $1\text{d}6 + 1$ damage, and a magical spear that deals 1d6 damage *with advantage.* Which magical weapons has a higher expected damage? *(Here, 1d6 means "roll one six-sided dice".)*